

MULTIMEDIATE '23: Engagement Estimation and Bodily Behaviour Recognition in Social Interactions

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ABSTRACT

Automatic analysis of human behaviour is a fundamental prerequisite for the creation of machines that can effectively interact with and support humans in social interactions. In MULTIMEDIATE '23, we address two key human social behaviour analysis tasks for the first time in a controlled challenge: engagement estimation and bodily behaviour recognition in social interactions. This paper describes the MULTIMEDIATE '23 challenge and presents novel sets of annotations for both tasks. For engagement estimation we collected novel annotations on the NOvice eXpert Interaction (NOXI) database. For bodily behaviour recognition, we annotated test recordings of the MPIIGroupInteraction corpus with the BBSI annotation scheme. In addition, we present baseline results for both challenge tasks.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

challenge, dataset, engagement, nonverbal behaviour

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1 INTRODUCTION

Artificial mediators [45], i.e. interactive intelligent agents that actively engage in a conversation in a human-like way have the potential to positively influence the course and/or outcomes of human interactions. They have been studied in a variety of contexts, including collaborative teamwork [10, 54], mental health [8], and education [16, 29]. A central prerequisite for effective and context-aware artificial mediation is the ability to comprehensively detect- and interpret the diverse set of social signals expressed by humans. At present, this challenge is still largely unsolved, and research on artificial mediators often has to rely on Wizard-of-Oz paradigms [8, 16, 29, 42, 51, 56].

With the multi-year MULTIMEDIATE challenge we contribute to realising the vision of autonomous artificial mediators by facilitating measurable advances on central conversational behaviour sensing and analysis tasks. The first iteration of the challenge in 2021 [37] has addressed eye contact detection and next speaker prediction while MULTIMEDIATE '22 has focused on backchannel analysis [1, 36]. In two separate tracks, MULTIMEDIATE '23 addresses the recognition of complex bodily behaviours, as well as the estimation of a persons' engagement level. Bodily behaviours such as fumbling, folded arms, or gesturing are a key social signal and were shown to

be connected to many important high-level phenomena including stress regulation, attraction, or social verticality [13, 22, 33, 57]. As a result, accurate recognition of bodily behaviours can serve as a building block for the recognition of such more abstract phenomena. Knowing how engaged participants are, individually or as a group, is important for a mediator whose goal it is to keep engagement at a high level. Engagement is closely linked to the previous MULTIMEDIATE tasks of eye contact detection [41, 46] as well as backchanneling [20].

With MULTIMEDIATE '23 we present the first challenge on engagement estimation and the recognition of bodily behaviours in social interaction. We define the tasks and evaluation criteria and describe new annotations collected on the NOVICE eXpert Interaction (NOXI) database [11], as well as on unreleased test recordings of MPIIGroupInteraction [39]. Furthermore, we present baseline approaches for both challenge tasks and report evaluation results. We make all collected annotations, baseline implementations, and raw feature representations publicly available for further use, even beyond the scope of MULTIMEDIATE '23.¹

2 RELATED WORK

We review previous works on methods and datasets for engagement estimation and bodily behaviour recognition in social interaction.

2.1 Engagement Estimation

Engagement has been investigated from various research angles, e.g. how to define, annotate, or to automatically predict it. Rich et al. [48] introduced a module for the recognition of engagement in human-robot interaction based on backchannels. Sanghvi et al. [50] predicted engagement based on body posture features. Bednarik et al. [6] focused on recognizing conversational engagement with gaze data. Research in detecting engagement in students is prolific and promising [19, 25]. Engagement is also often studied in children [47] and, more particularly, in children interacting with an artificial agent [24, 40, 44]. Guhan et al. [21] researched engagement in mental health patients, based on videos of the patient. Some datasets also offer engagement ratings, such as RECOLA [49], MHRI [14], and [23] with annotations from [6]. In Table 1 we provide an overview over the existing social interaction datasets with engagement annotations. The NOXI dataset annotated for MULTIMEDIATE '23 is significantly larger compared to previous datasets.

2.2 Bodily Behaviour Recognition

Bodily behaviours are key signals in social interactions and are related to many higher-level attributes. For example, displacement behaviours (e.g. fumbling, face-touching, or grooming) are associated with anxiety and stress regulation [5, 33, 34]. Leaning towards the interlocutor is connected with rapport [53] and crossed arms can be indicative of emotion expressions [60]. Further connections were found between bodily behaviours and liking [31, 32], attractiveness [57], and social verticality [22].

Despite this importance, little previous work addressed the recognition of bodily behaviours like fumbling, grooming, crossed arms, or gesturing in social interactions [3, 27]. While impressive progress was made on body- and hand pose estimation [12, 55], it is not

¹<https://multimEDIATE-challenge.org>

Corpus	Screen	Group size	Length	Part.
Guhan et al. [21]	✓	2	1h5m	13
RECOLA [49]	✓	2	3h50m	46
Bednarik et al. [6]	✓	4-7	6h	9 groups
MMHRI [14]	✗	2	6h	18
NOXI (ours)	✓	2	25h	87

Table 1: Social interaction datasets with engagement annotations, excluding MOOC and school settings and children as participants. *Screen* indicates whether interaction was screen-mediated, *Group size* the number of humans per interaction, *Length* the total duration of interactions, and *Part.* the total number of human participants.

a trivial task to establish the connection between low-level key-point detections and complex bodily behaviours that are relevant to the interaction. Furthermore, only a limited number of bodily behaviour recognition datasets containing spontaneous behaviour in social interactions is available. The PAVIS Face-Touching dataset [7] consists of a single annotated behaviour (face touching) in group discussions. The iMiGUE dataset [27] contains annotations of 32 behaviour classes annotated for speakers at sports press conferences. For the purpose of MULTIMEDIATE, the recently published BBSI dataset [3] is most relevant, which consists of 15 behaviour classes annotated for all participants of 3-4 person group conversations. Such group conversations are one of the main application domains of artificial mediators.

3 CHALLENGE DESCRIPTION

In the following we present the two challenge tasks and the utilised datasets. For both tasks test samples (without ground truth) are released to participants before the challenge deadline. Participants in turn submit their predictions for evaluation.

3.1 Engagement Estimation Task

Task definition. The task includes the continuous, frame-wise prediction of the level of conversational engagement of each participant on a continuous scale from 0 (lowest) to 1 (highest). Participants are encouraged to investigate multimodal as well as reciprocal behaviour of both interlocutors in the Novice-Expert Interaction corpus. We make use of the Concordance Correlation Coefficient (CCC) [26] to evaluate predictions on the test set.

Dataset. The NOVICE eXpert Interaction (NOXI) database [11] is a corpus of dyadic, screen-mediated face-to-face interactions in an expert-novice knowledge sharing context. In a session, one participant assumes the role of an expert and the other participant the role of a novice. Figure 1 shows two users during interaction. NOXI includes interactions recorded at three locations (France, Germany and UK), spoken in eight languages (English, French, German, Spanish, Indonesian, Arabic, Dutch and Italian), discussing a wide range of topics. The dataset offers over 25 hours (x2) of recordings of dyadic interactions in natural settings, featuring synchronized audio, video (25fps), and motion capture data (using a Kinect 2.0).

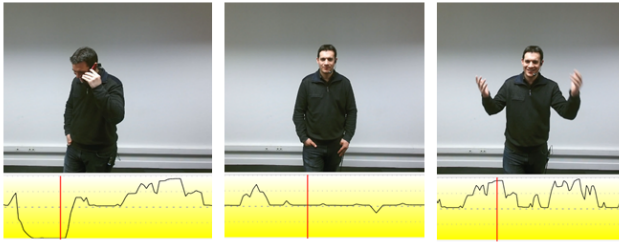


Figure 1: Snapshots of scenes of a participant in the NOXI corpus being disengaged (left), neutral (center) and highly engaged (right).

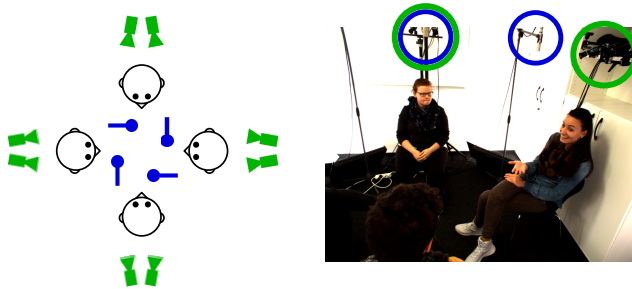


Figure 2: Setup of the MPIIGroupInteraction dataset. Reproduced with permission from the authors of [39].

We will use subset of this corpus containing 48 sessions for training and 16 sessions for testing (75/25 split). We aimed to obtain data of spontaneous behavior in a natural setting on a variety of discussion topics. Therefore, one of the main design goals was to match recorded participants based on their common interests. This means that we first gathered potential experts willing to share their knowledge about one or more topics they were knowledgeable and passionate about, and secondly we recruited novices willing to discuss or learn more about the available set of topics offered by experts. The corpus further introduces interruptions of the novices in order to provoke experts' reactions when conversational engagement gets interrupted. In particular, for this challenge, each session has been annotated in a continuous matter, meaning each video frame has a score between 0 and 1. Each rating was performed by at least two (up to 7) annotators (Average: 3.6 raters per session). We created gold standard annotations by calculating the mean over all raters. The NOXI dataset can be obtained from the website².

3.2 Bodily Behaviour Recognition Task

Task definition. We formulate bodily behaviour recognition as a multi-label classification task. Challenge participants are required to predict which of 15 behaviour classes are present in a 64 (2.13 sec) frame input window. For each 64-frame window, we provide a frontal view on the target participant, as well as two side views (left and right). As the behaviour classes on this task are highly

unbalanced, we will measure performance using average precision computed per class and aggregated using macro averaging, i.e. giving the same weight to each class. This encourages challenge competitors to develop novel methods to improve performance on challenging low-frequency classes.

Dataset. As in MULTIMEDIATE '21 [37], our challenge is based on the MPIIGroupInteraction dataset [38, 39]. This dataset has served as a basis for diverse tasks, including emergent leadership detection [35], eye contact detection [18, 30, 38], next speaker prediction [9], backchannel analysis [1, 52], and body language detection [3]. The MPIIGroupInteraction corpus consists of 22 group discussions between three to four people, each lasting for 20 minutes [39]. This year's bodily behaviour task is based on the recently collected BBSI annotations [3], consisting of 15 bodily behaviour classes annotated on the whole MPIIGroupInteraction corpus. For MULTIMEDIATE '23, we excluded "Lean towards" as inter-annotator agreement was reported to be very low on this class. We collected bodily behaviour annotations for the remaining 14 classes on 996 samples obtained from six unpublished test recordings of MPIIGroupInteraction following the BBSI protocol [3]. To reach high-quality annotations on the test set, we obtained consensus decisions from three annotators. All classes except the "Stretching" class were present on the test set. The MPIIGroupInteraction dataset can be obtained from the website³.

4 EXPERIMENTS AND RESULTS

We are providing a baseline model for each task. This section describes the training methodology as well as the utilized features and results achieved for both tasks.

4.1 Engagement Estimation

4.1.1 Approach. For the engagement estimation task we rely on a set of multimodal features comprising body posture, facial features and vocal features, followed by a fully connected neural network with three hidden layers of size 112 each. To prevent overfitting we rely on a dropout layer after the second hidden layer with a dropout rate of 0.25. The network has been trained using the Adam optimizer and the mean squared error loss function. All hyperparameters have been optimized using the hyperband search algorithm of the KerasTuner framework [43].

Head Features. We extracted features from participants' head and face using OpenFace 2.0 [4]. All features were extracted for each video frame. The resulting feature vectors are consisting of 68 3D facial landmarks, 56 3D eye landmarks, presence and intensity of 18 action units as well as markers for detection success, detection certainty facial position and rotation. Furthermore, we also use 17 action units provided by the Microsoft Kinect sensor.

Pose Features. We extract body pose estimates using OpenPose [12] as well as the Microsoft Kinect sensor data, resulting in the estimation of 350 data points comprising information about the location of various joints as well as their rotation.

Voice Features. For the paralinguistic assessment of engagement we extracted two feature sets over a one-second sliding window with a stride of 40ms to match the frame rate of the video stream. The

²https://multimediate-challenge.org/datasets/Dataset_NoXi/

³https://multimediate-challenge.org/datasets/Dataset_MPII/

Features	Val CCC	Test CCC
<i>Head</i>		
openface	0.23	0.21
AUs	0.31	0.22
<i>Body</i>		
skeleton	0.47	0.43
openpose	0.53	0.43
<i>Voice</i>		
gemaps	0.58	0.55
soundnet	0.54	0.49
<i>Multimodal</i>		
feature fusion + pca	0.71	0.59

Table 2: Concordance correlation coefficient (CCC) of our baseline on engagement detection validation and test sets.

first feature set is the Geneva Minimalistic Acoustic Parameter Set (eGeMAPS) [17]. This set consists of 54 acoustic parameters that are commonly applied to tasks like depression, mood, and emotion recognition [58]. Secondly, we used pretrained version of Soundnet [2] to extract sound embeddings from the raw signal. Soundnet is a deep convolutional neural network that has already been shown to provide effective features for vocal social signal analysis [59].

In our baseline approach, we fused the feature vectors of all modalities into one feature vector. As a large number of features can lead to overfitting we applied a PCA, reducing the number of features to 83 principal components.

4.1.2 Results. The results are depicted in Table 2. Among the single modalities the vocal features are clearly outperforming the body and head features on the validation set as well as on the test set. However, the multimodal feature fusion shows that the combination of all features still outperforms just using vocal features substantially. The additional value added by head and body features indicates that the expression of engagement is not clearly bound to one modality but should be analyzed considering multiple modalities.

4.2 Bodily Behaviour Recognition

4.2.1 Approach. As our baseline solution, we chose the Video Swin Transformer [28], which produced recent state-of-the-art results in action recognition tasks. It operates on fixed inputs of length 32 frames and size of 224×224 pixels. Given the input videos of length 64 frames and of larger resolutions, we set the stride to 2, that is we took every second frame, and we resized the video accordingly. We assigned input clips with multiple corresponding behavior class labels and clips of different viewpoints are treated as independent samples during training. To the clips with no labels, we assigned a new behavior class called *Background*, and, instead of the 14, trained the model in a 15-class multi-label setup. To aggregate predictions across views at test time, we averaged the scores obtained from all three views. We used the Swin Base model that is pre-trained on ImageNet and Kinetics-400, and fine-tuned it on the MPIIGroupInteraction dataset for only one epoch with learning rate 10^{-3} and

Approach	Val MAP	Test MAP
random baseline	0.0884	0.2355
w/o bkgd class, frontal view	0.3974	0.5315
w/o bkgd class, side view 1	0.3030	0.4341
w/o bkgd class, side view 2	0.3628	0.4893
w/o bkgd class, max of views	0.4087	0.5333
w/o bkgd class, mean of views	0.4084	0.5402
w/ bkgd class, frontal view	0.4051	0.5498
w/ bkgd class, side view 1	0.3096	0.4451
w/ bkgd class, side view 2	0.3686	0.4641
w/ bkgd class, max of views	0.4062	0.5443
w/ bkgd class, mean of views	0.4099	0.5628

Table 3: Validation and test results for the random baseline and different variants of the Video Swin Transformer.

with AdamW optimizer. Our implementation uses the open-source toolbox MMAction2 [15] built on top of PyCharm.

4.2.2 Results. Results of multiple ablations are reported in Table 3. We evaluated our approach against ablations that operate on single views, against an aggregation strategy using the maximum across views, as well as against not using an additional background class during training. The best mean average precision (MAP) on both validation and test sets was achieved by averaging across views and training with a background class. While the inclusion of the background class only led to minor improvements, averaging across views yielded consistent improvements. The best single view was the frontal view, and side views resulted in a significant performance drop. All results clearly outperformed the random baseline. Results on the test set tend to be systematically higher, likely as a result of the higher quality annotations, and the lack of the “Stretching” class on the test set which as a result is always evaluated with 1.

5 CONCLUSION

We introduced MULTIMEDIATE '23, the first challenge addressing engagement estimation and bodily behaviour recognition in social interactions in well-defined conditions. We presented publicly available datasets and evaluation protocols for both tasks, and evaluated baseline approaches. The evaluation server will remain accessible to researchers even beyond the MULTIMEDIATE challenge, contributing to continuing progress on both tasks.

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